

Lab Report for Interview 1

Programming Lab: ASCII Display Program

Create a menu driven program that allows the user to select between displaying uppercase letters, lowercase letters, or exiting the program using numbers (for example, 1 for uppercase, 2 for lowercase, and 3 for exit). The program should include functions to collect the following:

1. **Uppercase Letters** (Call DisplayUppercaseASCII() to display uppercase letters (A-Z) with ASCII values)
2. **Lowercase Letters** (Call DisplayLowercaseASCII() to display lowercase letters (a-z) with ASCII values)

Using this information, the program calculates and displays the following:

- Line by line values for letters and their associated ASCII values
-

The program should adhere to the following input validation:

- Do not accept negative numbers
- Do not accept characters
- Screenshots must contain relevant information about the case you are demonstrating and documenting.
- Your lab report must look professional and complete. Demonstrate your program is robust, reliable, scalable

Three files must be submitted:

- A PDF document of the lab report (ASCII.pdf)
- The C++ source file (ASCII.cpp)

System Requirements

What will the program do and how?

The program will display characters and their associated ASCII values. First, it will display a Main Menu where the user may select one of the following operations:

- Display Uppercase Letters (A = 65, B = 66, ..., Z = 90)
- Display Lowercase Letters (a = 97, b = 98, ..., z = 122)
- Exit the program

The program will then prompt the user to enter a valid number (e.g., from 1 to 3).

System Analysis

Document the data flow from the input to the output (IPO)

Input

- **Menu choices:**
 - For DisplayUppercaseASCII() (1)
 - For DisplayLowercaseASCII() (2)
 - To exit the program (3)

Processing

- **Control flow:**
 - **Startup:** Enter main() and prompt for menu choice:
 - Uppercase Letters (A-Z) → DisplayUppercaseASCII()
 - Lowercase Letters (a-z) → DisplayLowercaseASCII()
 - X exits process (exit(0))
- **Data Management:** Store values in a fixed array of Category structs (size = NUM_CATEGORIES (10))
- **Validation:** Integers only range between 1 and 3

Output

- **Menu:** Title, Main Menu (print once), short prompt: Enter choice (1-3)
- **Exit:** Goodbye! Exiting program... when 3 is chosen.

System Design

Describe the logic that will solve the problem. The logic must be directly implemented in C++.

Modularity

Describe the functions that will be implemented within the algorithm.

The DisplayUppercaseASCII() function loops through the ASCII codes for uppercase letters (65 to 90) and converts it to its character equivalent.

```
1 // 2. Display uppercase letters (A-Z) with ASCII values
2 void DisplayUppercaseASCII() {
3     for (char c = 'A'; c <= 'Z'; c++) {
4         cout << c << " = " << int(c) << endl;
5     }
6 }
```

Figure 1 DisplayUppercaseASCII() method from ASCII.cpp

The DisplayLowercaseASCII() function loops through the ASCII codes for lowercase letters (97 to 122) and converts it to its character equivalent.

```
1 // 3. Display lowercase letters (a-z) with ASCII values
2 void DisplayLowercaseASCII() {
3     for (char c = 'a'; c <= 'z'; c++) {
4         cout << c << " = " << int(c) << endl;
5     }
6 }
```

Figure 2 DisplayLowercaseASCII() method from ASCII.cpp

Main Function and Logic

Summarize the main design and logic.

main() prints the menu and loops until the user enters a range between 1 and 3. Then main() calls the while(choice != 3) loop and iterates until the user exits the program.

```
1  #include <iostream>
2  using namespace std;
3
4  void DisplayUppercaseASCII();
5  void DisplayLowercaseASCII();
6
7  // 1. Program execution starts here
8  int main() {
9      // 1.1 Declare and initialize choice
10     int choice = 0;
11
12     // 1.2 Loop until user chooses EXIT
13     while (choice != 3) {
14         // 1.3 Display menu
15         cout << "\n==== ASCII MENU =====\n";
16         cout << "1. Uppercase Letters (A-Z)\n";
17         cout << "2. Lowercase Letters (a-z)\n";
18         cout << "3. Exit Program\n";
19
20         // 1.4 Get user input
21         cout << "\nEnter Choice: ";
22         cin >> choice;
23
24         // 1.5 Call functions or exit program
25         switch (choice) {
26             case 1: DisplayUppercaseASCII(); break;
27             case 2: DisplayLowercaseASCII(); break;
28             case 3: cout << "Exiting program...\n"; break;
29             default: cout << "ERROR: Invalid Input - Enter a
digit (1-3)...\n";
30         }
31     }
32
33     return 0;
34 }
```

Figure 3 main() method from ASCII.cpp

Implementation

Describe the program line by line via comments.

***View attached documentation for Implementation...**

Testing

Data flow from the input to the output (IPO).

```
===== ASCII MENU =====  
1. Uppercase Letters (A-Z)  
2. Lowercase Letters (a-z)  
3. Exit Program  
  
Enter Choice: 1  
A = 65  
B = 66  
C = 67  
D = 68  
E = 69  
F = 70  
G = 71  
H = 72  
I = 73  
J = 74  
K = 75  
L = 76  
M = 77  
N = 78  
O = 79  
P = 80  
Q = 81  
R = 82  
S = 83  
T = 84  
U = 85  
V = 86  
W = 87  
X = 88  
Y = 89  
Z = 90
```

Figure 4 Compiler displays the main menu once and prompts the user to enter a choice. User enters 1 and compiler displays uppercase letters and associated ASCII value

```
Enter Choice: 2
a = 97
b = 98
c = 99
d = 100
e = 101
f = 102
g = 103
h = 104
i = 105
j = 106
k = 107
l = 108
m = 109
n = 110
o = 111
p = 112
q = 113
r = 114
s = 115
t = 116
u = 117
v = 118
w = 119
x = 120
y = 121
z = 122
```

Figure 5 User enters 2 and compiler displays lowercase letters and associated ASCII value

```
Enter Choice: 4
ERROR: Invalid Input - Enter a digit (1-3)...
```

Figure 6 User enters a character and a number outside of the range and the program gives an error message

```
Enter Choice: 3
Exiting program...
```

Figure 7 User enters 2 and compiler displays exit message as the program successfully exits